

Current-algebra results in the path-integral framework

R. E. Gamboa Saravi and F. A. Schaposnik

*Departamento de Física, Facultad de Ciencias Exactas, Universidad Nacional de La Plata,
C.C. No. 67, 1900 La Plata, Argentina*

J. E. Solomin

*Departamento de Matemática, Facultad de Ciencias Exactas, Universidad Nacional de La Plata,
C.C. No. 67, 1900 La Plata, Argentina*

(Received 26 September 1985)

A systematic way of getting current-algebra results while working in the path-integral formulation of quantum field theory is presented. The method is employed in the analysis of a two-dimensional model where fermions are coupled to a non-Abelian gauge field, leading to the equal-time commutation relations for currents and energy-momentum tensor. The anomalous commutators show that the Schwinger terms contain covariant derivatives related to the nontriviality of fermion Jacobians arising from chiral transformations. The relation with the nonlinear σ -model results is discussed.

I. INTRODUCTION AND RESULTS

Since the basic ideas of current algebra were introduced by Gell-Mann¹ in 1961, the subject has found an increasing number of applications both from the theoretical and the experimental sides.² An important result, already obtained in 1969,³ was related to the fact that current commutators were not independent of interactions, because of the appearance of anomalies related to the singularity structure of operator products for small separations.

The problem of anomalies then grew as a subject by itself, being related to the renormalizability of gauge theories,⁴⁻⁶ to the construction of composite models,⁷ low-energy effective Lagrangians,⁸ etc. It has also been actively investigated in connection with supersymmetry,⁹ physical theories in higher-dimensional spaces,¹⁰ and, more recently, after the discovery of a new kind of anomaly,¹¹ in the selection of string theories based on the requirement that gravitational and gauge anomalies be absent.^{12,13}

In this last context, interest in constructing the commutator algebra of currents and the energy-momentum tensor of two-dimensional field theories (such as nonlinear σ model) has been stressed^{14,15} as a way of providing in a very simple way representations of Kac-Moody and Virasoro algebras which can then be used to analyze the spectrum of candidate string theories. In these investigations, the non-Abelian bosonization scheme developed by Witten¹⁶ for free-fermion models is a basic tool.

Now, non-Abelian bosonization in the operator framework has not yet led to the complete solution of interacting fermion models such as, for example, two-dimensional QCD, as happens for the Abelian ones. There is however an alternative approach based on the path-integral formulation of quantum field theory which has shown to be very useful in the solution of both Abelian and non-Abelian two-dimensional models.¹⁷⁻¹⁹ In this approach, the role of anomalies in bosonization is transparent and

the complete solution of models such as the Gross-Neveu model or QCD₂ can be envisaged (including the computation of correlation functions, which is beyond the power of the Bethe-ansatz approach usually employed in solving these models¹⁹).

In the path-integral framework, the operator bosonization recipe is replaced by a chiral change in the fermion variables and the anomaly appears as the associated nontrivial Jacobian; one has still to develop in this context a systematic method of discovering the signal of anomalous commutators of the kind leading, for example, to a Kac-Moody algebra, as happens in free-fermion models. As first pointed out by Fujikawa,²⁰ it is precisely the nontriviality of fermion Jacobians which is responsible, in the path-integral approach, for anomalous commutators appearing in the operator language. Jacobians are then the objects to look for if one intends to discover whether algebras of the Kac-Moody type also arise in models like QCD₂.

Based on these ideas, we present in this work a systematic way of getting current-algebra results while working in the path-integral framework. We also apply our method to the analysis of a model where two-dimensional fermions couple to a background non-Abelian gauge field, having in mind the resolution of QCD₂. Our main results are the following: the necessity of regularizing fermion determinants introduces anomalies under certain transformations which transmit to currents and energy-momentum tensor (evaluated in this approach as functional derivatives of the corresponding fermion determinant). The anomalous behavior of commutators can then be easily derived from the study of the transformation of currents and energy-momentum tensor under the (quantum) conflictive symmetries. As an example, we consider the behavior of vector currents under chiral rotations, being the result, in this case, Eq. (3.15).

When applied to fermions coupled to non-Abelian gauge fields in two-dimensional space-time, our method

easily leads to the equal-time commutator relations for currents [Eq. (4.22)] and for the energy-momentum tensor [Eqs. (4.54), (4.56) and (4.57)]. As we discussed in an earlier work where the technique presented here was briefly outlined,²¹ the Schwinger term appearing in anomalous current commutators contains in this case covariant (and not ordinary) δ -function derivatives and hence it does not lead to an ordinary Kac-Moody algebra but to an extension of it. It is possible, however, to transform the currents in such a way that they satisfy the ordinary algebra [Eqs. (4.25) and (4.26)]. Because of this fact, the energy-momentum tensor algebra is formally the same as in the free model, except for the fact that here, because of the anomaly, currents are not conserved [however, the relation between energy-momentum tensor T_{++} (T_{--}) and currents J_+ (J_-) is still of the form $T_{++} \sim J_+^2$ ($T_{--} \sim J_-^2$), as is shown in Eqs. (4.32)].

The outline of the paper is as follows. In Sec. II we establish our conventions and review the way anomalous Ward-Takahashi identities are obtained in the path-integral formalism. We then describe in Sec. III how current-algebra results are derived in this framework. Finally, we apply in Sec. IV our method to the case of a two-dimensional model of fermions interacting with an external gauge field, in two-dimensional space-time.

II. THE GENERAL SETTING

We review in this section some by now very well-known results concerning the path-integral derivation of (anomalous) Ward-Takahashi identities associated with symmetries of the Lagrangian defining the dynamics of some fermionic model. In particular, it will become clear how anomalies arise, as first stressed by Fujikawa,²² due to the necessity of regularizing the fermion determinant (or, in other words, the necessity of defining adequately the path-integral fermionic measure).

The basic tool in this functional approach is the generating functional Z , written in terms of the Lagrangian and, possibly, of the external sources. We shall be mainly interested in models with charged fermions interacting with non-Abelian gauge fields, which will be considered as classical background fields but which can be, of course, quantized (after the addition of the $F_{\mu\nu}^2$ term). Z reads

$$Z = Z[A_\mu, g_{\mu\nu}; \bar{\eta}, \eta] \equiv \int \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{iS[\bar{\psi}, \psi, A] + iS_{\text{sources}}}, \quad (2.1)$$

where the (Minkowski) action S is

$$S = \int d\mu_x [\bar{\psi}(x)(i\partial + i\mathbf{F} + e\mathbf{A})\psi(x)], \quad (2.2)$$

and the source term is given by

$$S_{\text{sources}} = \int d\mu_x [\bar{\psi}(x)\eta(x) + \bar{\eta}(x)\psi(x)]. \quad (2.3)$$

We consider Dirac fermions in the fundamental representation of some symmetry group [in the examples we discuss, it will be taken as $SU(N)$], interacting with the gauge field A_μ taking values in the corresponding Lie algebra. We also include a background gravitational field $g_{\mu\nu}(x)$ in order to compute the energy-momentum tensor from the generating functional. At the end of the calculations,

we shall take the flat-space limit. We then have

$$d\mu_x = d^d x \sqrt{g}, \quad (2.4)$$

$$\sqrt{g} = +\sqrt{\det g_{\mu\nu}}. \quad (2.5)$$

We use curved-space-time γ matrices denoted by $\underline{\gamma}_\mu(x)$ satisfying

$$\underline{\gamma}_\mu(x)\underline{\gamma}_\nu(x) + \underline{\gamma}_\nu(x)\underline{\gamma}_\mu(x) = 2g_{\mu\nu}(x). \quad (2.6)$$

These matrices are related to the flat-space ones by means of the vierbeins:

$$\underline{\gamma}_\mu(x) = b_\mu^a(x)\gamma_a \quad (2.7)$$

with

$$g_{\mu\nu}(x) = b_\mu^a(x)b_\nu^b(x)\eta_{ab}, \quad (2.8)$$

where η_{ab} is the flat-space-time metric in Minkowski space. The spinorial affine connection $\Gamma_\mu(x)$ is defined by

$$\Gamma_\mu(x) = \frac{1}{4}\gamma_a\gamma_b b_\lambda^a(x)g^{\lambda\sigma}(x)\nabla_\mu b_\sigma^b(x) \quad (2.9)$$

with

$$\nabla_\mu b_\sigma^a = \partial_\mu b_\sigma^a - \Gamma_{\mu\sigma}^\lambda b_\lambda^a, \quad (2.10)$$

where $\Gamma_{\mu\nu}^\lambda$ is the normal Christoffel symbol. $\bar{\psi}$ is defined through

$$\bar{\psi} = \psi^\dagger \gamma_0 \quad (2.11)$$

and $A_\mu = A_\mu^{k,l}$ takes values in the Lie algebra which in the examples will be taken as that of $SU(N)$, with generators t^k satisfying

$$[t^k, t^l] = if^{klm}t^m, \quad \text{tr} t^k t^l = \delta^{kl}. \quad (2.12)$$

Equations (2.2) being quadratic, the path integral (2.1) can be evaluated in closed form:

$$Z[A, g; \bar{\eta}, \eta] = \int \det(i\partial + i\mathbf{F} + e\mathbf{A}) \times \exp \left[\int \bar{\eta}(x) G_A(x, y) \eta(y) d\mu_x d\mu_y \right] \quad (2.13)$$

with the Green's function G_A defined through the equation

$$[i\partial + i\mathbf{F} + e\mathbf{A}(x)]G_A(x, y) = \frac{1}{\sqrt{g}}\delta(x - y). \quad (2.14)$$

Concerning the determinant in (2.13), it will be meaningful only after some regularization is prescribed, since the product of the eigenvalues of the differential operator $\mathcal{D}$,

$$\mathcal{D} = i\partial + i\mathbf{F} + e\mathbf{A}, \quad (2.15)$$

increases without bound. We shall adopt throughout this paper the ζ -function regularization:

$$\det \mathcal{D} \equiv \exp \left[-\frac{d\zeta}{ds}(s; \mathcal{D}) \right] \Big|_{s=0}. \quad (2.16)$$

A general definition of ζ for elliptic invertible operators $\mathcal{D}$

of order $m > 0$ defined on a compact manifold M , of dimension d (space-time) without boundary, can be given in the form²³

$$\zeta(s; D) = \int d\mu_x \text{tr} K_{-s}(D; x, x). \quad (2.17)$$

Here, tr means the matricial trace and $K_{-s}(D; x, x)$ is the meromorphic extension of the evaluation of the kernel of D^{-s} on the diagonal $x = y$.

Once a regularization is adopted, one has to determine if it is compatible with all the original symmetries of the theory. One of the reasons the ζ -function method is particularly useful lies in the observation that it is, *per se*, gauge invariant and can be used even when D is non-Hermitian (for example, when γ_5 couplings are present in the Lagrangian) (see Ref. 23).

Other symmetries of the theory may be spoiled by the regularization procedure. Indeed, consider some (continuous) symmetry transformation leaving invariant the action

$$\begin{aligned} \psi &\rightarrow \psi' = \psi(x) + \delta_\alpha \psi, \\ \bar{\psi} &\rightarrow \bar{\psi}' = \bar{\psi}(x) + \delta_\alpha \bar{\psi}, \\ S[A, \bar{\psi}, \psi] &= S[A, \bar{\psi}', \psi'] \end{aligned} \quad (2.18)$$

(we have considered a transformation depending on an infinitesimal parameter α). At the classical level there is an associated conserved current (which can be derived via the Noether theorem) corresponding either to an internal or to a space-time symmetry. Equivalently, there is a classically conserved quantity, the associated charge.

In the usual operator approach to the quantum theory, one studies at this stage the commutator algebra for these currents and charges and determines if they correspond to the ones derived by naive manipulation of the canonical commutation relations for the fundamental fields. In this framework, it is sufficient to consider the transformation (2.18) as depending on an infinitesimal global (i.e., constant) parameter.

On the contrary, in the path-integral approach, one promotes (2.18) to a local transformation and write $\alpha = \alpha(x)$. The action is, in general, noninvariant under this transformation:

$$\psi \rightarrow \psi' = \psi + \delta_{\alpha(x)} \psi, \quad \bar{\psi} \rightarrow \bar{\psi}' = \bar{\psi} + \delta_{\alpha(x)} \bar{\psi}, \quad (2.19)$$

$$S \rightarrow S' = S + \int d\mu_x \Lambda_\alpha(x). \quad (2.20)$$

Now, since the transformation (2.18) corresponds to a symmetry of the theory, we can write Λ as the divergence of some current. As an example, for the case of a non-Abelian symmetry, with $\alpha = \alpha^a t^a$ one gets

$$\Lambda_\alpha(x) = -(D_\mu^{ab} J^{b\mu}) \alpha^a(x). \quad (2.21)$$

In order to study the possible quantum conservation of this current, that is, to derive the corresponding Ward-Takahashi identity, one just takes (2.19) as a change of variables to be performed in the generating functional (2.1) (with sources taken to be 0 for simplicity). In his pioneering work²² Fujikawa stressed that one should expect, *a priori*, a nontrivial associated Jacobian,

$$\mathcal{D}\bar{\psi}' \mathcal{D}\psi' = J(\alpha) \mathcal{D}\bar{\psi} \mathcal{D}\psi, \quad (2.22)$$

each time the operator approach leads to an anomaly, since it is in the fermion measure where the quantum aspect is taken into account in the path-integral formulation. Within the ζ -function regularization method, this Jacobian can be understood as the factor relating the fermionic determinants before and after the transformation (2.19) (Ref. 23)

$$Z = \det(\mathcal{D}) = J(\alpha) \det(D_\alpha), \quad (2.23)$$

where D_α is defined from (2.20) by writing

$$S' = \int \bar{\psi}' D_\alpha \psi' d\mu_x. \quad (2.24)$$

In this language the “classical” transformation law

$$\mathcal{D} \rightarrow D_\alpha \quad \text{when } \psi \rightarrow \psi + \delta_\alpha \psi, \quad \bar{\psi} \rightarrow \bar{\psi} + \delta_\alpha \bar{\psi} \quad (2.25)$$

becomes

$$\det(\mathcal{D}) \rightarrow J(\alpha) \det(D_\alpha) \quad (2.26)$$

at the quantum level; that is, there is a phase $[\ln J(\alpha)]$ picked up in the quantum treatment of the symmetry transformation. At this point one can make contact with some recent work by Faddeev.²⁴ Consider, for example, the case of an infinitesimal chiral transformation. In this case (2.18) can be written as

$$\psi' = \mathcal{U}(\alpha) \psi, \quad \bar{\psi}' = \bar{\psi} \mathcal{U}(\alpha), \quad (2.27)$$

with

$$\mathcal{U}(\alpha) = e^{i\gamma_5 \alpha} \simeq 1 + i\gamma_5 \alpha.$$

Once one arrives at Eq. (2.26), one can restrict α to be constant and then, the theory being chiral invariant,

$$D_\alpha = e^{i\gamma_5 \alpha} \mathcal{D} e^{i\gamma_5 \alpha} = \mathcal{D}. \quad (2.28)$$

However,

$$\det(e^{i\gamma_5 \alpha} \mathcal{D} e^{i\gamma_5 \alpha}) = e^{iQ[\alpha]} \det(\mathcal{D})$$

with $Q[\alpha]$ related to the topological charge of the background field A_μ . (Note that no contradiction arises since, when $Q[\alpha] \neq 0$, there are necessarily zero modes and then $\det(\mathcal{D}) = 0$. If zero modes are carefully taken into account, one then obtains the usual index result (see Ref. 25).)

Now, coming back to the Ward-Takahashi identity associated with the transformation (2.19), one has to take account of the change in the action, Eqs. (2.20) or (2.24), and also in the fermionic measure, given by the Jacobian $J(\alpha)$ defined by Eq. (2.23). After the transformation, the generating functional reads

$$\begin{aligned} Z = J(\alpha) \mathcal{N} \int \mathcal{D}\bar{\psi}' \mathcal{D}\psi' \exp(iS'[\bar{\psi}', \psi', A]) \\ \times \exp \left[i \int d\mu_x \Lambda_\alpha(x) \right]. \end{aligned} \quad (2.29)$$

Since Z cannot depend on α ,

$$\frac{\delta Z}{\delta \alpha} = 0 = i \left\langle \frac{\delta \int d\mu_x \Lambda_\alpha(x)}{\delta \alpha(x)} \right\rangle + \frac{\delta \ln J(\alpha)}{\delta \alpha(x)}. \quad (2.30)$$

Of course, if (2.21) represents a gauge transformation,

$$\psi \rightarrow \psi + it^a \delta \alpha^a \psi, \quad \bar{\psi} \rightarrow \bar{\psi} - \bar{\psi} i t^a \delta \alpha^a, \quad (2.31)$$

the ζ -function regularization being gauge invariant, from Eq. (2.23) one has $J(\alpha) = 1$. On the other, $\Lambda_a(x)$ is given by Eq. (2.21) and then Eq. (2.30) leads to the conservation of the vector current at the quantum level:

$$\langle D_\mu^{ab} J_\mu^b \rangle = 0. \quad (2.32)$$

If instead of Eq. (2.31) one now considers a chiral rotation as in Eq. (2.27),

$$\psi \rightarrow \psi + i \gamma_5 \alpha(x) \psi, \quad \bar{\psi} \rightarrow \bar{\psi} + \bar{\psi} i \gamma_5 \alpha(x), \quad (2.33)$$

one then has

$$J(\alpha) = \exp \left[\frac{i}{16\pi^2} \int \alpha(x) \text{tr}^* F_{\mu\nu} F^{\mu\nu} d\mu_x \right]. \quad (2.34)$$

Concerning the change in the action, one easily finds

$$\Lambda(x) = -i \alpha(x) \partial_\mu (\bar{\psi} \gamma_\mu \gamma_5 \psi) \quad (2.35)$$

and, hence, at the quantum level one has

$$\langle \partial_\mu (\bar{\psi} \gamma_\mu \gamma_5 \psi) \rangle = \frac{i}{16\pi^2} \text{tr}^* F_{\mu\nu} F^{\mu\nu} \quad (2.36)$$

(note that A_μ plays the role of a background field). Of course, one can consider also non-Abelian chiral rotations and study the corresponding Ward-Takahashi identities. Within the ζ -function method, one obtains in this case the so-called gauge-invariant anomaly, which differs from the consistent (gauge-variant) one. One can derive one anomaly from the other by modifying the current definition, as explained in Ref. 26. It will be useful for further purposes to quote the result for $d=2$ space-time:

$$\begin{aligned} D_\mu^{ab} \mathcal{J}_\mu^{sb} &= \frac{ie}{2\pi} \epsilon^{\mu\nu} F_{\mu\nu}^a \quad \text{gauge-invariant anomaly,} \\ D_\mu^{ab} \mathcal{J}_\mu^{sb} &= \frac{ie}{2\pi} \epsilon^{\mu\nu} \partial_\mu A_\nu^a \quad \text{consistent anomaly.} \end{aligned} \quad (2.37)$$

III. COMMUTATOR ALGEBRA FROM PATH-INTEGRAL RESULTS

We describe in this section how current-algebra results can be derived using the path-integral approach.^{21,27} In particular, the relation between anomalous commutators and the careful regularization of the path-integral-generating functional (i.e., the fermion determinant) is clarified, supporting the association of anomalies with Jacobian factors in the path-integral framework, as first discussed by Fujikawa.²²

Once the generating functional (2.12) is properly defined (i.e., the fermion determinant is adequately regularized), any correlation function can be obtained just by functional differentiation with respect to the sources. As an example, the two-point function reads

$$\langle \psi(x) \bar{\psi}(y) \rangle \equiv - \frac{1}{Z} \frac{\delta^2 Z}{\delta \bar{\eta}(x) \delta \eta(y)} \Big|_{\eta=\bar{\eta}=0} = G_A(x, y). \quad (3.1)$$

Moreover, because of the presence of the background fields $(A_\mu, g_{\mu\nu})$ fermion currents and the energy-

momentum tensor can be evaluated with *no need of any new regularization*²⁸ from (2.13) (with $\eta = \bar{\eta} = 0$):

$$J_\mu = \langle \bar{\psi}(x) \gamma_\mu t^a \psi(x) \rangle = \frac{-i \delta \ln \det(\mathcal{D})}{\delta A_\mu^a(x)}, \quad (3.2)$$

$$T_{\mu\nu} = \left\langle \bar{\psi}(x) \frac{i}{2} \gamma_\mu \tilde{\sigma}_{\nu\lambda} \psi(x) \right\rangle = 2i \frac{\delta \ln \det(\mathcal{D})}{\delta g_{\mu\nu}(x)}. \quad (3.3)$$

Now, if one intends to calculate the product of currents at the same point, new singularities appear and one does need an additional regularization prescription. Exactly as in the operator approach, one can remedy this problem by defining, for example,

$$\langle J_\mu^a(x) J_\nu^b(y) \rangle_{\text{reg}} \equiv [\langle J_\mu^a(x) J_\nu^b(y) \rangle + \Delta J]_{x=y}, \quad (3.4)$$

choosing ΔJ in (3.4) so as to render the left-hand side finite. We shall show this prescription at work in the next section, in the discussion of the QCD₂ current algebra.

How does one infer from the path-integral results [i.e., from vacuum expectation values (VEV's) of currents expressed as path integrals] the algebra of currents? Interest in this problem has been recently renewed by the discovery of a remarkable connection between dual models and certain field theories, leading, for example, to the representation of Kac-Moody and Virasoro algebras by starting from the nonlinear σ model.^{14-16,19,29} Since the equal-time commutator control quantum symmetry transformations, one should expect that nontrivial results will show up in the path-integral approach through nontrivial Jacobians related to these transformations.^{20,21}

Suppose one has found a classically conserved current K_μ associated with some classical symmetry transformation:

$$\psi \rightarrow \psi'_\alpha = (1 + i\alpha)\psi, \quad \bar{\psi} \rightarrow \bar{\psi}'_\alpha = \bar{\psi}(1 + i\bar{\alpha}) \quad (3.5)$$

with $\bar{\alpha} = -\alpha$, $\bar{\alpha} = \gamma_5 \alpha$, etc. One then constructs the conserved charge

$$Q[\alpha] = \int d^2x K_0(x) \alpha(x)$$

and, from it, one introduces in the operator framework the unitary operator e^{iQ} , which is the generator of the symmetry transformations

$$e^{-iQ[\alpha]} \psi(x) e^{iQ[\alpha]} = e^{i\alpha} \psi, \quad (3.6)$$

as can be very easily established from Eq. (3.5) and the canonical commutation relation for the ψ fields:

$$\{\psi_\alpha^\dagger(x), \psi_\beta(y)\}_{\text{ET}} = \delta_{\alpha\beta} \delta(x-y). \quad (3.7)$$

The action of the operator on some other current $J_\mu = J_\mu[\bar{\psi}; \psi]$ reads

$$J_\mu = e^{-iQ[\alpha]} J_\mu e^{iQ[\alpha]} = J_\mu - i[Q, J_\mu], \quad (3.8)$$

and hence the change in the current components under the associated symmetry transformation reads

$$\delta_\alpha J'_\mu(x) = -i[Q_\alpha, J_\mu(x)]. \quad (3.9)$$

In the path-integral approach one can also analyze the action of this symmetry transformation by studying VEV's of currents. Indeed, one can write, for the transformed current,

$$\langle J'_\mu \rangle = \frac{1}{\det[\mathcal{D}(A)]} \int \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{iS[\bar{\psi}, \psi, A]} J_\mu[\bar{\psi}', \psi'_\alpha] \quad (3.10)$$

with $\psi'_\alpha, \bar{\psi}'_\alpha$ given by (3.5). The corresponding inverse transformation is

$$\psi = (1 - i\alpha)\psi'_\alpha, \quad \bar{\psi} = \bar{\psi}'_\alpha(1 - i\bar{\alpha}). \quad (3.11)$$

One can now consider Eq. (3.11) as a change of variables and try to express (3.10) in terms of $\psi'_\alpha, \bar{\psi}'_\alpha$. Of course, a nontrivial Jacobian should be present in principle and hence we write

$$\langle J'_\mu \rangle = \frac{J(\alpha)}{\det[\mathcal{D}(A)]} \int \mathcal{D}\bar{\psi}'_\alpha \mathcal{D}\psi'_\alpha e^{iS_\alpha} J_\mu[\bar{\psi}', \psi'_\alpha], \quad (3.12)$$

$$S_\alpha = S[\bar{\psi}'_\alpha(1 - i\bar{\alpha}), (1 - i\alpha)\psi'_\alpha, A]. \quad (3.13)$$

That is, the VEV of the transformed current can be computed as a path integral of the original current but with an S_α transformed action. Introducing the normalization constant $\mathcal{N}_\alpha$ defined as

$$\mathcal{N}_\alpha = \int \mathcal{D}\bar{\psi}'_\alpha \mathcal{D}\psi'_\alpha e^{iS_\alpha}, \quad (3.14)$$

we have

$$\langle J'_\mu \rangle = \frac{J(\alpha)}{\det[\mathcal{D}(A)]} \mathcal{N}_\alpha \langle J_\mu \rangle_\alpha, \quad (3.15)$$

where we indicate with $\langle \rangle_\alpha$ that the VEV is to be computed with the transformed action S_α .

As we shall see in the next section, relation (3.15) can be very easily used when the symmetry associated with the charge (3.6) is strongly related to the solubility of the model under study. In this way, commutation relations can be very easily established from the path-integral results when studying two-dimensional fermion models where chiral symmetry plays a central role.

IV. THE WORLD OF CURRENTS IN TWO-DIMENSIONAL FERMION MODELS

We apply in this section the method described above in order to solve a model where fermions interact with a background gauge field in two space-time dimensions, in the way of the theory of currents, namely, by deriving the bosonized form of the fermionic currents and then expressing the energy-momentum tensor in terms of it; in this way, we get the commutator algebra starting from path-integral results.

We are considering an action of the form (2.2) but in two-dimensional space-time. We first consider the flat-space case and take (3.4) as giving infinitesimal chiral rotations:

$$\psi' = (1 + i\gamma_5 \delta\alpha)\psi, \quad \bar{\psi}' = \bar{\psi}(1 + i\gamma_5 \delta\alpha). \quad (4.1)$$

Now, because of the peculiar properties of two-dimensional γ matrices, the transformed action (3.13) in this case reads

$$S_\alpha[\bar{\psi}'_\alpha, \psi'_\alpha, A, \alpha] = S[\bar{\psi}'_\alpha, \psi'_\alpha, A_\alpha], \quad (4.2)$$

where

$$A_\alpha = A - e\gamma_\mu \gamma_5 D^\mu \delta\alpha, \quad (4.3)$$

$$D_\mu = \partial_\mu - ie[A_\mu, \quad]. \quad (4.4)$$

That is, the transformed action can be written as the original one but in terms of a “transformed” gauge field (because of the relation $\gamma^\mu \gamma_5 = i\epsilon^{\mu\nu} \gamma_\nu$, one is sure $A_{\alpha\mu}$ is indeed a gauge field taking values in the Lie algebra of the symmetry group).

In the path-integral approach, it is convenient to take the gauge field in the so-called decoupling gauge, where it can be written in terms of a Lie-algebra-valued scalar:¹⁷

$$A(x) = -\frac{i}{e}(\partial e^{i\gamma_5 \phi})e^{-i\gamma_5 \phi}. \quad (4.5)$$

Then, A_α takes the form

$$A_\alpha(x) = -\frac{i}{e}(\partial e^{i\gamma_5(\phi - \delta\alpha)})e^{-i\gamma_5(\phi - \delta\alpha)}. \quad (4.6)$$

We have taken $[\phi, \delta\alpha] = 0$ since, the vector currents being gauge covariant, one can always reduce the chiral rotation to one parallel to ϕ by an adequate gauge transformation without losing generality. Now, inserting (4.2)–(4.6) in (3.13), the normalization $\mathcal{N}_\alpha$ reads

$$\mathcal{N} = \det[\mathcal{D}(A_\alpha)] \quad (4.7)$$

and hence, from (3.15) we have, for the fermion vector current,

$$J_\mu^a = \bar{\psi}\gamma_\mu t^a \psi, \quad (4.8)$$

$$\langle J_\mu^a \rangle = \left[\frac{J(\alpha) \det[\mathcal{D}(A_\alpha)]}{\det[\mathcal{D}(A)]} \right] \langle J_\mu^a(A_\alpha) \rangle.$$

Now, the term in large parentheses on the right-hand side (RHS) of (4.8) can be shown to be 1 since

$$\det[\mathcal{D}(A)] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{iS[\bar{\psi}, \psi, A]}$$

$$= J(\alpha) \int \mathcal{D}\bar{\psi}'_\alpha \mathcal{D}\psi'_\alpha e^{iS[\bar{\psi}'_\alpha, \psi'_\alpha, A_\alpha]}$$

or

$$\det[\mathcal{D}(A)] = J(\alpha) \det[\mathcal{D}(A_\alpha)]. \quad (4.9)$$

We have, then,

$$\langle J_\mu^a(A) \rangle = \langle J_\mu^a(A_\alpha) \rangle. \quad (4.10)$$

That is, the VEV of the chiral transformed current in the background A_μ coincides with the VEV of the current in the transformed background $A_{\alpha\mu}$. (From here on we shall omit the $\langle \rangle$ symbol except for the cases when confusion can arise.)

Now, the point is one can compute the RHS in (4.10) very simply within the path-integral approach.³⁰ Indeed, the fermion determinant for massless QCD₂ is known to be¹⁷

$$W[A] = \ln \det[\mathcal{D}(A)]$$

$$= \frac{1}{8\pi} \int d\mu_x \text{tr}[\partial^\mu \mathcal{U}^{-2}(\phi, 1) \partial_\mu \mathcal{U}^2(\phi, 1)] - \frac{1}{4\pi} \int d\mu_x \int_0^1 d\alpha \frac{\epsilon^{\mu\nu}}{\sqrt{g}} \text{tr}(\mathcal{U}^{-2} \partial_\mu \mathcal{U}^2 \mathcal{U}^{-2} \partial_\nu \mathcal{U}^2 \mathcal{U}^{-2} \partial_\alpha \mathcal{U}^2), \quad (4.11)$$

where

$$\mathcal{U}(\phi, \alpha) = e^{i\alpha\phi(x)}, \quad \alpha \in [0, 1]. \quad (4.12)$$

Using expression (3.2) one can easily compute the exact VEV of J_μ which can be compactly written in terms of light-cone variables as³⁰

$$J_+ = -\frac{i}{4\pi} \mathcal{U}^2 D_+ \mathcal{U}^{-2} = -\frac{i}{4\pi} \mathcal{U}^{-1} (\partial_+ \mathcal{U}^2) \mathcal{U}^{-1}, \quad (4.13)$$

$$J_- = -\frac{i}{4\pi} \mathcal{U}^{-2} D_- \mathcal{U}^2 = -\frac{i}{4\pi} \mathcal{U}^{-1} (\partial_- \mathcal{U}^2) \mathcal{U}^{-1},$$

where

$$x_\pm = x^0 \pm x^1, \quad J_\pm = J^0 \pm J^1, \quad (4.14)$$

$$D_\pm = \partial_\pm - ie[A_\pm,], \quad \mathcal{U} = e^{i\phi}.$$

The current J_μ is covariantly conserved,

$$D_+ J_- + D_- J_+ = 0, \quad (4.15)$$

while the axial one,

$$J_\mu^{5a} = \langle \bar{\psi} \gamma_\mu \gamma_5 t^a \rangle, \quad (4.16)$$

is anomalous,

$$D_+ J_-^5 + D_- J_+^5 = A \quad (4.17)$$

with A given by Eq. (2.37):

$$A^a = -\frac{ie}{2\pi} \epsilon^{\mu\nu} F_{\mu\nu}^a. \quad (4.18)$$

Using (4.6) and (4.11) we can now explicitly evaluate the chiral transformed vector current (4.10):

$$J'_\pm(A) = J_\pm(A) \mp \left[\frac{1}{2\pi} D_\pm \delta\alpha + i[J_\pm, \delta\alpha] \right]. \quad (4.19)$$

One can infer from this equation (relating the VEV of fermion currents) the commutator of currents with the chiral charge, since in the operator approach we have [Eq. (3.9)]

$$\delta_{Q_5} J_\pm = -i[Q_5, J_\pm] \quad (4.20)$$

with

$$Q_5 = \int dx J_0^5(x, t) \delta\alpha(x) = \int dx J_1(x, t) \delta\alpha. \quad (4.21)$$

In order to make Eq. (4.19) compatible with Eq. (4.20), one necessarily has

$$[J_\pm^a(x, t), J_\pm^b(y, t)] = f^{abc} J_\pm^c(x, t) \delta(x - y) + \frac{i}{2\pi} D_\pm^{ab} \delta(x - y), \quad (4.22)$$

$$[J_+^a(x, t), J_-^b(y, t)] = 0.$$

The second term on the RHS of Eq. (4.22) is the so-called Schwinger term (which in two dimensions is finite). It can also be evaluated by means of the Bjorken-Johnson-Low limit.³¹ (See Ref. 2 for a very clear discussion and additional references as well as Ref. 32 for a more recent discussion of the commutator anomaly in the perturbative approach.)

It is important to stress that, instead of the usual δ function, the background field induces a covariant derivative in current commutators (4.22), a phenomenon also present in other models^{24,3}. Because of the chiral anomaly, the currents are not covariantly conserved separately, but

$$D_+ J_- = -D_- J_+ = \frac{A}{2} \quad (4.23)$$

with $A = A^a t^a$ given by (4.18).

Only in the free case ($e=0$) do the commutation relations (4.22) take the usual form—that of a Kac-Moody algebra in an infinite spatial interval. For $e \neq 0$ instead of the central extension with an integer k , there is a background field dependence in the Schwinger term. There exists, however, a unitary transformation leading from currents satisfying (4.22) to new ones $j_\pm$, giving the realization of a Kac-Moody algebra. Indeed, consider the currents $j_\pm$ defined as

$$j_+ = \mathcal{U} J_+ \mathcal{U}^{-1} = -\frac{i}{4\pi} (\partial_+ \mathcal{U}^2) \mathcal{U}^{-2}, \quad (4.24)$$

$$j_- = \mathcal{U}^{-1} J_- \mathcal{U} = -\frac{i}{4\pi} (\partial_- \mathcal{U}^{-2}) \mathcal{U}^2.$$

It is easy to show that the commutation relations for them take the form

$$[j_+^a(x, t), j_+^b(y, t)] = 2f^{abc} j_+^c(x, t) \delta(x - y) + \frac{i}{2\pi} \delta^{ab} \partial_+(x - y),$$

$$[j_-^a(x, t), j_-^b(y, t)] = 2f^{abc} j_-^c(x, t) \delta(x - y) + \frac{i}{2\pi} \delta^{ab} \partial_-(x - y), \quad (4.25)$$

$$[j_+^a(x, t), j_-^b(y, t)] = 0.$$

It must be stressed that these currents differ from those corresponding to the free-fermion model since they do not satisfy relations of the form $\partial_+ j_- = \partial_- j_+ = 0$ and, hence, one cannot infer from this that $j_+ = j_+(x^+)$, $j_- = j_-(x^-)$.

Now, because of the fact they obey the algebra (4.25), we shall be able to derive the corresponding algebra satisfied by the energy-momentum tensor since, as we shall see, it depends on traces of products of currents and hence it does not distinguish between $J_\pm$ and $j_\pm$.

It is interesting to discuss at this point the symmetry

behind (4.25) and its connection with the chiral transformation we have performed. Consider the transformation

$$\mathcal{U}^2 \rightarrow \Omega_+ \mathcal{U}^2 \Omega_- , \quad (4.26)$$

where $\Omega_+ = \Omega_+(x^+)$ and $\Omega_- = \Omega_-(x^-)$ are matrices belonging to the gauge group G , satisfying

$$[\mathcal{U}, \Omega_{\pm}] = 0 . \quad (4.27)$$

The group of transformations (4.26) corresponds, in the bosonized form of the model we are studying, to the chiral-symmetry transformations we have been discussing in the fermion language, i.e., transformation (4.1). Note that we have defined the transformation (4.26) for $\mathcal{U}^2$, the relevant variable appearing in the fermion determinant, Eq. (4.11), and not for $\mathcal{U}$, which originally appeared in the expression for the gauge field in the decoupling gauge, defined through the relations

$$A_+ = \frac{i}{e} \mathcal{U}^{-1} \partial_+ \mathcal{U}, \quad A_- = -\frac{i}{e} \partial_- \mathcal{U} \mathcal{U}^{-1} . \quad (4.28)$$

If $\Omega_{\pm}$ are taken as infinitesimal transformations

$$\Omega_{\pm} = I + 2\delta\alpha_{\pm}^a t^a , \quad (4.29)$$

then it is easy to show that the currents defined in (4.24) change like

$$j_{\pm} \rightarrow j'_{\pm} = j_{\pm} \mp \left[\frac{1}{2\pi} D_{\pm} \delta\alpha_{\pm} + 2i[j_{\pm}, \delta\alpha_{\pm}] \right] , \quad (4.30)$$

which shows that $j_{\pm}$ are the generators of the group $G(x_{\pm})$ which represents the Kac-Moody algebra with central extension $k=1$. There is, however, an important difference between our result and that obtained for free fermions (or equivalently for the nonlinear σ model with the Wess-Zumino term). In the present case, because of the presence of the background gauge field, currents $j_{\pm}$ are not conserved since (4.11) is not the action for the theory but just a part of it (having in mind QCD₂, one should include the $F_{\mu\nu}^2$ term and also the Faddeev-Popov determinant for the decoupling gauge).

The next step in our investigation of anomalous commutators will be the evaluation of the energy-momentum tensor using (3.3) and (4.11). One very easily gets

$$\langle T_{++} \rangle = 2\pi \text{tr} \langle J_+^2 \rangle, \quad \langle T_{--} \rangle = 2\pi \text{tr} \langle J_-^2 \rangle , \quad (4.31)$$

where we have defined

$$\langle T_{++} \rangle = \langle T_{00} \rangle + \langle T_{01} \rangle, \quad \langle T_{--} \rangle = \langle T_{00} \rangle - \langle T_{01} \rangle . \quad (4.32)$$

As we observed in the preceding sections, we have obtained a finite result because of the fact that we have adopted a regularization that allows differentiation with no need of new subtractions. The result can be interpreted as the operator level as the definition of a normal ordering

$$T_{\pm\pm} = 2\pi \text{tr} : J_{\pm}^2 : \quad (4.33)$$

or, equivalently, to a point-splitting regularization procedure. Indeed, from our result we can define

$$T_{++}^{\text{reg}} = 2\pi \text{tr} J_+ J_+ + \Delta T \equiv 2\pi : J_+^2 : \quad (4.34)$$

or

$$\langle T_{++}^{\text{reg}} \rangle = 2\pi \text{tr} \langle J_+^2 \rangle + \Delta T . \quad (4.35)$$

Now,

$$\begin{aligned} \langle J_+^2(x) \rangle &= \frac{1}{Z} \frac{\delta^2 Z}{\delta A_+^2(x)} \\ &= \frac{\delta}{\delta A_+(x)} \left[\frac{\delta}{\delta A_+(x)} \ln Z \right] + \left[\frac{\delta \ln Z}{\delta A_+(x)} \right]^2 , \end{aligned} \quad (4.36)$$

and hence

$$\langle J_+^2 \rangle = \langle J_+ \rangle^2 + \frac{\delta \langle J_+(x) \rangle}{\delta A_+(x)} . \quad (4.37)$$

Comparing (4.33) and (4.37) we get

$$\Delta T = -2\pi \frac{\delta \langle J_+^a(x) \rangle}{\delta A_+^a(x)} , \quad (4.38)$$

which shows the singularity manifest here as a $\delta(0)$. It is interesting to note that the quadratic dependence of the energy-momentum tensor with the fermion currents is satisfied in this case and in the free-fermion model.^{14,15} The result suggests this model can be investigated in the way of the theory of currents, since the energy-momentum tensor, which in principle determines the dynamics, can be expressed in terms of currents only.

Using Eq. (4.33) we can now compute the commutator between T_{++} (T_{--}) and J_+ (J_-) following the same lines as before. Indeed,

$$\langle T_{++} \rangle = \frac{1}{\det[\mathcal{D}(A)]} \int \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{iS[\bar{\psi}, \psi, A]} T_{++} \quad (4.39)$$

with T_{++} on the RHS given by the (classical) expression

$$T_{++} = \frac{1}{2} \bar{\psi} \gamma_+ D_+ \psi . \quad (4.40)$$

In order to derive the commutation relations between T_{++} and J_+ , we consider a combined chiral and gauge transformation since

$$\delta_Q T_{++} + \delta_Q T_{++} = -i \int \delta\alpha^a(x) [J_+^a, T_{++}] d\mu_x . \quad (4.41)$$

Now, because of gauge invariance, $\delta_Q \langle T_{++} \rangle = 0$. Concerning the chiral change,

$$\psi \rightarrow \psi' = e^{i\delta\alpha\gamma_5} \psi, \quad \bar{\psi} \rightarrow \bar{\psi}' = \bar{\psi} e^{i\delta\alpha\gamma_5} , \quad (4.42)$$

$$\begin{aligned} &\frac{1}{2} \langle \bar{\psi}' \delta_+ D_+ (A) \psi' \rangle \\ &= \frac{1}{\det[\mathcal{D}(A)]} \int \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{iS[\bar{\psi}, \psi, A]} \bar{\psi}' \gamma_+ D_+ \psi' . \end{aligned} \quad (4.43)$$

Performing the inverse transformation as a change of variables and proceeding as before,

$$\langle T'_{++}(A) \rangle = \frac{J(\delta\alpha)}{\det[\mathcal{D}(A)]} \int \mathcal{D}\bar{\psi}' \mathcal{D}\psi' \bar{\psi}' \gamma_+ D_+(A) \times \psi' e^{iS[\bar{\psi}', \psi', A']} . \quad (4.44)$$

Now, from the identity

$$\bar{\psi}' \gamma_+ D_+(A) \psi' = \bar{\psi}' \gamma_+ D_+(A') \psi' + e J_+^a (A_+^a - A_+'^a) , \quad (4.45)$$

we have

$$\langle T'_{++}(A) \rangle = J(\delta\alpha) \frac{\det[\mathcal{D}(A')]}{\det[\mathcal{D}(A)]} \times [\langle T_{++}(A') \rangle + e \langle J_+^a(x) \rangle (A_+^a - A_+'^a)] \quad (4.46)$$

or, using (4.9),

$$\langle T'_{++}(A) \rangle = \langle T_{++}(A') \rangle + e \langle J_+^a(A) \rangle (A_+^a - A_+'^a) . \quad (4.47)$$

But, from (4.3),

$$A_+^a - A_+'^a = \frac{1}{e} D_+^{ab} \delta\alpha^b \quad (4.48)$$

and, hence,

$$\langle T_{++}(A') \rangle = -2 \langle J_+^a(A) \rangle D_+^{ab} \delta\alpha^b + \langle T_{++}(A) \rangle + \langle J_+^a(A) \rangle D_+^{ab} \delta\alpha^b . \quad (4.49)$$

Then,

$$\delta \langle T_{++}(A) \rangle = (-2 + 1) \langle J_+^a(A) \rangle D_+^{ab} \delta\alpha^b , \quad (4.50)$$

this corresponding to the commutation relation

$$[J_+^b(x', t), T_{++}(x, t)] = -i J_+^a(x, t) D_+^{ab} \delta(x' - x) , \quad (4.51)$$

from which the correctly normalized relation follows,

$$\int [J_+^b(x', t), T_{++}(x, t)] dx' = i D_+^{ab} J_+^a , \quad (4.52)$$

showing that our path-integral method gives a result consistent with the appropriate normal ordering needed in the operator approach. Again, the ordinary derivative of the δ function appearing in the free-model case^{14,15} is here substituted for by a covariant derivative. One can recover the free-model result for the currents defined in (4.24). One easily gets, from (4.52),

$$\delta \langle T_{++} \rangle = \mathcal{U} \langle J_+^a \rangle \mathcal{U}^{-1} D_+^{ab} \mathcal{U} \delta\alpha = - \langle j_+ \rangle \partial_+ \delta^{ab} \delta\alpha , \quad (4.53)$$

and since

$$T_{++} = 2\pi \text{tr} J_+^2 = 2\pi \text{tr} j_+^2 , \quad (4.54)$$

the relation (4.51) becomes, in terms of j_+ ,

$$[j_+^a(x', t), T_{++}(x, t)] = -i j_+^a(x, t) \partial_+ \delta(x' - x) . \quad (4.55)$$

In order to evaluate the commutation relations between energy-momentum tensor components, one should proceed as before, by studying the change in $T_{\mu\nu}$ under conformal transformations. This is not necessary in the present case since the fact that one can express $T_{\pm\pm}$ in terms of currents satisfying the algebra (4.25) ensures one should get the ordinary commutation relations, i.e.,

$$[T_{++}(x, t), T_{++}(y, t)] = 2i [J_+(x, t), J_+(y, t)] \delta'(x - y) - \frac{i}{6\pi} \delta'''(x - y) , \quad (4.56)$$

$$[T_{--}(x, t), T_{--}(y, t)] = 2i [J_-(x, t), J_-(y, t)] \delta'(x - y)$$

$$- \frac{i}{6\pi} \delta'''(x - y) .$$

ACKNOWLEDGMENTS

We are grateful to C. M. Naón for useful discussions. This work was supported in part by Consejo Nacional de Investigaciones Científicas y Técnicas, CIC, and SECYT, Argentina. R.E.G.S. and F.A.S. were supported by CIC, Buenos Aires, Argentina. J.E.S. was supported by CONICET, Argentina.

¹M. Gell-Mann, Phys. Rev. **125**, 1067 (1962); Physics **1**, 63 (1964), and references therein.

²See, for example, S. Treiman, R. Jackiw, and D. Gross, *Lectures on Current Algebra and Its Applications* (Princeton University Press, Princeton, 1972).

³S. Adler, Phys. Rev. **177**, 2426 (1969); J. S. Bell and R. Jackiw, Nuovo Cimento **60A**, 67 (1969).

⁴D. Gross and R. Jackiw, Phys. Rev. D **6**, 477 (1972).

⁵H. Georgi and S. Glashow, Phys. Rev. D **6**, 429 (1972).

⁶C. Bouchiat, J. Iliopoulos, and Ph. Meyer, Phys. Lett. **38B**, 519 (1972).

⁷G. 't Hooft, in *Recent Developments in Gauge Theories* (Plenum, New York, 1980).

⁸E. Witten, Nucl. Phys. **B156**, 269 (1979); G. Veneziano, *ibid.*

B171, 253 (1980).

⁹S. Ferrara and B. Zumino, Nucl. Phys. **B87**, 207 (1975).

¹⁰P. H. Frampton, Phys. Lett. **122B**, 351 (1983).

¹¹L. Alvarez Gaumé and E. Witten, Nucl. Phys. **B234**, 269 (1984).

¹²M. B. Green and J. Schwarz, Phys. Lett. **149B**, 117 (1984).

¹³D. Gross, J. Harvey, E. Martinec, and R. Rohm, Phys. Rev. Lett. **54**, 502 (1985).

¹⁴D. Nemeschansky and S. Yankielowicz, Phys. Rev. Lett. **54**, 620 (1985).

¹⁵S. Antoniadis and C. Bachas, SLAC Report No. 3625, 1985 (unpublished).

¹⁶E. Witten, Commun. Math. Phys. **92**, 455 (1984).

¹⁷R. E. Gamboa Saraví, F. A. Schaposnik, and J. E. Solomin,

- Nucl. Phys. **B185**, 239 (1981); Phys. Rev. D **30**, 1353 (1984).
- ¹⁸K. Furuya, R. E. Gamboa Saraví, and F. A. Schaposnik, Nucl. Phys. **B208**, 159 (1982).
- ¹⁹A. M. Polyakov and P. B. Wiegmann, Phys. Lett. **131B**, 121 (1983); **141B**, 223 (1984).
- ²⁰K. Fujikawa, Phys. Lett. **108B**, 33 (1982).
- ²¹R. E. Gamboa Saraví, C. M. Naón, and F. A. Schaposnik, Phys. Lett. **163B**, 213 (1985).
- ²²K. Fujikawa, Phys. Rev. Lett. **42**, 1195 (1979); Phys. Rev. D **21**, 2848 (1980).
- ²³R. E. Gamboa Saraví, M. A. Muschietti, F. A. Schaposnik, and J. E. Solomin, Ann. Phys. (N.Y.) **157**, 360 (1985).
- ²⁴L. Faddeev, Phys. Lett. **145B**, 81 (1984).
- ²⁵R. E. Gamboa Saraví, M. A. Muschietti, and J. E. Solomin, Commun. Math. Phys. **93**, 407 (1984).
- ²⁶W. Bardeen and B. Zumino, Nucl. Phys. **B244**, 421 (1984).
- ²⁷H. Aratyn and P. H. Damgaard, Nucl. Phys. **B241**, 445 (1984).
- ²⁸R. E. Gamboa Saraví, M. A. Muschietti, F. A. Schaposnik, and J. E. Solomin, J. Math. Phys. **26**, 2045 (1985).
- ²⁹V. G. Knizhnik and A. B. Zamolodchikov, Nucl. Phys. **B247**, 83 (1984).
- ³⁰R. E. Gamboa Saraví, C. M. Naón, and F. A. Schaposnik, Phys. Lett. **153B**, 97 (1985).
- ³¹J. D. Bjorken, Phys. Rev. **148**, 1467 (1966); K. Johnson and F. E. Low, Prog. Theor. Phys. Suppl. **37-38**, 74 (1966).
- ³²M. Kahagashi, K. Seo, and A. Sugamoto, KEK Report No. KEK TH-112, 1985 (unpublished).
- ³³H. J. de Vega, H. Eichenherr, and J. M. Maillet, Phys. Lett. **132B**, 337 (1983); H. J. de Vega, LP THE Paris Report No. LP THE 85/07, 1985 (unpublished).